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INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CRUX

It connects people who have problems with students who can solve them.

- Team Name: COGNIHUB
- Theme: Open Innovation
- PS Category: Software
- Team ID: DEX26297



THE PROBLEM

Real problems and student builders rarely meet

Farmers, clinics, shopkeepers, NGOs and colleges all have problems they can't solve alone.

Students want real problems to work on. Today the two sides rarely find each other.

The gap

Student teams have no structured, evidence-backed way to find a real problem and there is no place for people with real problem to share.

1 People with problems don't know who can help

NGOs, clinics and colleges often need a software/hardware fix but can't reach student builders.

2 Students guess instead of solving real needs

Teams spend days picking an idea, often one that is already solved or that nobody asked for.

3 Nobody checks if it's worth building

Nothing tells a team whether an idea is original or whether they can finish it in time.

OUR IDEA: CRUX

1

Discover

A live feed of real problems from news, open government data and public grievances, sorted by sector and place.

2

Choose

Student teams browse by sector and place and pick the problem they want to take on.

3

Score

Rates how realistic the idea is for your time, team size and tech stack.

4

Match

Looks at what your team can build (web, app, AI, design) and suggests problems that fit.

5

Brief

Turns your pick into a clear, submission-ready problem statement.

How it answers the problem

Instead of guessing, a team gets evidence: where the problem is, whether it's taken, and whether we can really build it in time. People with problem get free help from motivated students.

What makes it different

Most idea generators just hand out topics. Crux starts from real, current problems, posted by the people It scores your ideas and give a problem statement for students to solve.

TECHNICAL APPROACH

Collect
News, open govt data, project archives

2
Clean and tag
Remove duplicates, tag sector and location

3
Analyse
Embeddings find similar existing solutions

4
Score and match
Novelty, feasibility and skill fit

5
Present
Feed, validator and brief generator

What we'll build it with

Front end
HTML with React and Tailwind CSS

Back end
Python with FastAPI

AI and search
Sentence embeddings for similarity, an LLM API for summaries and briefs

Data
PostgreSQL ; hosted on free-tier cloud

This is a design-stage proposal. Our first build would be Discover, Validate and Brief, with Score and Match following.

FEASIBILITY AND VIABILITY

Why this is doable

- The data we need is public and free to use
- Standard tools: React, Python and open-source AI models
- Six people can split it cleanly: UI, back end, data, AI, design and testing

Risk	Our plan
Why would people post problems?	Start with college,NGOs and local offices that already wants student help.
Fake and Low-quality post	A verified organisation tag and a report button for bad posts.
Scores that feel arbitrary	Show the reasoning behind each score and let teams adjust the weights
Too much to build at once	Build in phases: Discover, Validate and Brief first, Score and Match second

BUSINESS CANVAS

Mission and vision

Mission: Connect people who have problem with students who can solve them.

Vision: Every real problem finds a motivated team, and students learn by building what people actually need.

Market and competition

People with problems: NGOs, colleges, clinics, local offices and small businesses. Builders: students.

Idea generators and trend lists exist, but none start from real problems posted by the people who face them.

How it earns

Free for students and small NGOs.

Paid features for colleges and organisations that want private problem lists and progress dashboards. Providing priority notification.

What we need

Running costs are low: Free-tier hosting and open-source models.

The main spend is AI API calls and keeping the problem feed fresh.

IMPACT AND BENEFITS

People with Problems

- Free help from motivated student teams
- Problems that were too small or costly for professionals get attention
- A clear brief to hand to builders

Organisers and Colleges

- Fewer duplicate projects
- More original, better-scoped submissions
- A clear view of which problems students care about

Students

- Start building sooner
- Pick problems they can actually finish
- Learn to test an idea before betting on it, a skill that lasts beyond the hackathon

Our Goal: Every real problem finds a motivated team, and every team starts from a real need.

RESEARCH AND REFERENCES

What we're drawing on

- Open Government Data Platform India (data.gov.in) for public datasets
- Centralised Public Grievance Redress and Monitoring System, for the kinds of problems citizens report
- Hackathon archive of past problem statements
- User's problem statement.
- Reimers and Gurevych (2019), Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks
- Common research papers

Where we are

Concept stage: problem defined, solution and architecture planned. We've built a concept model of the website with sample data.

Next: Build the problem board, idea check , Discover, Validate and Brief, then publish the code on GitHub.